Track 2 report — from a confirmed genotype to graded, approved medicines, with its own negative control

Standing: draft, written 2026-09-19 from executed runs. Iteration 2 of Track 2 supersedes iteration 1 (never uploaded): the mechanism card is at **version 4** with all nine measured facts filled — the last one, the aneuploid fraction in the sequenced sample, by iteration 2's step 8 — and the report now tells the whole path, steps 1 to 8. Every number is read from a table with a provenance manifest; every claim was registered before its test; every cited paper was resolved against PubMed. **Not uploaded.** Track 2 accepts one submission (due about 2026-10-20).

Written for a panel that includes clinicians and patient advocates as well as scientists. Each technical term appears once with its meaning in brackets; the full dictionary is served at `/learn` on the dashboard, and the same tables render on its `/track2` page.

> The challenge's own words, kept in view throughout: *submissions are hypotheses for follow-up, not evidence that a medicine works.* Nothing below is a recommendation to treat.

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1. The answer in one page

The child. A boy with mosaic variegated aneuploidy (MVA: some of his cells carry a wrong number of chromosomes, and different cells are wrong in different ways), short stature, failure to thrive, calcium deposits in the kidneys, and a rhabdomyosarcoma (a childhood cancer of muscle tissue). Eight clinical terms were delivered with the case; no karyotype was.

The cause, found twice. Track 1 found two different broken copies of one gene, `BUB1B` — one from each parent (a *compound heterozygote*) — and the challenge's answer key confirmed the pair with a full match at rank 1. Iteration 2 then rebuilt the answer **blind**, from the raw reads and the eight clinical terms, by routes sharing no code with the first: the same pair came out first. The gene makes BubR1, the **brake** on cell division that holds a dividing cell until every chromosome is attached. One copy makes nothing (a *stop* change); the other makes a weakened protein (a *missense* change in the protein's last stretch). Less brake, not no brake — the only pattern compatible with life, and the pattern every published case shows.

Nine measured facts sit on the mechanism card so that no drug argument rests on the allele labels alone. The one that matters most for this report: **no chromosome in the sequenced sample carries a mosaic population above 2–3 % of cells** (step 8, at 80 % power), where the statistic used earlier would have found nothing at any fraction. That is a statement about the sequenced sample — its tissue is not named by the delivery, and a karyotype's fraction is measured in cultured cells — not about the diagnosis.

Four ways a drug could act, graded on a five-rung ladder written before the search, then run again with the gene swapped for two decoys as a control:

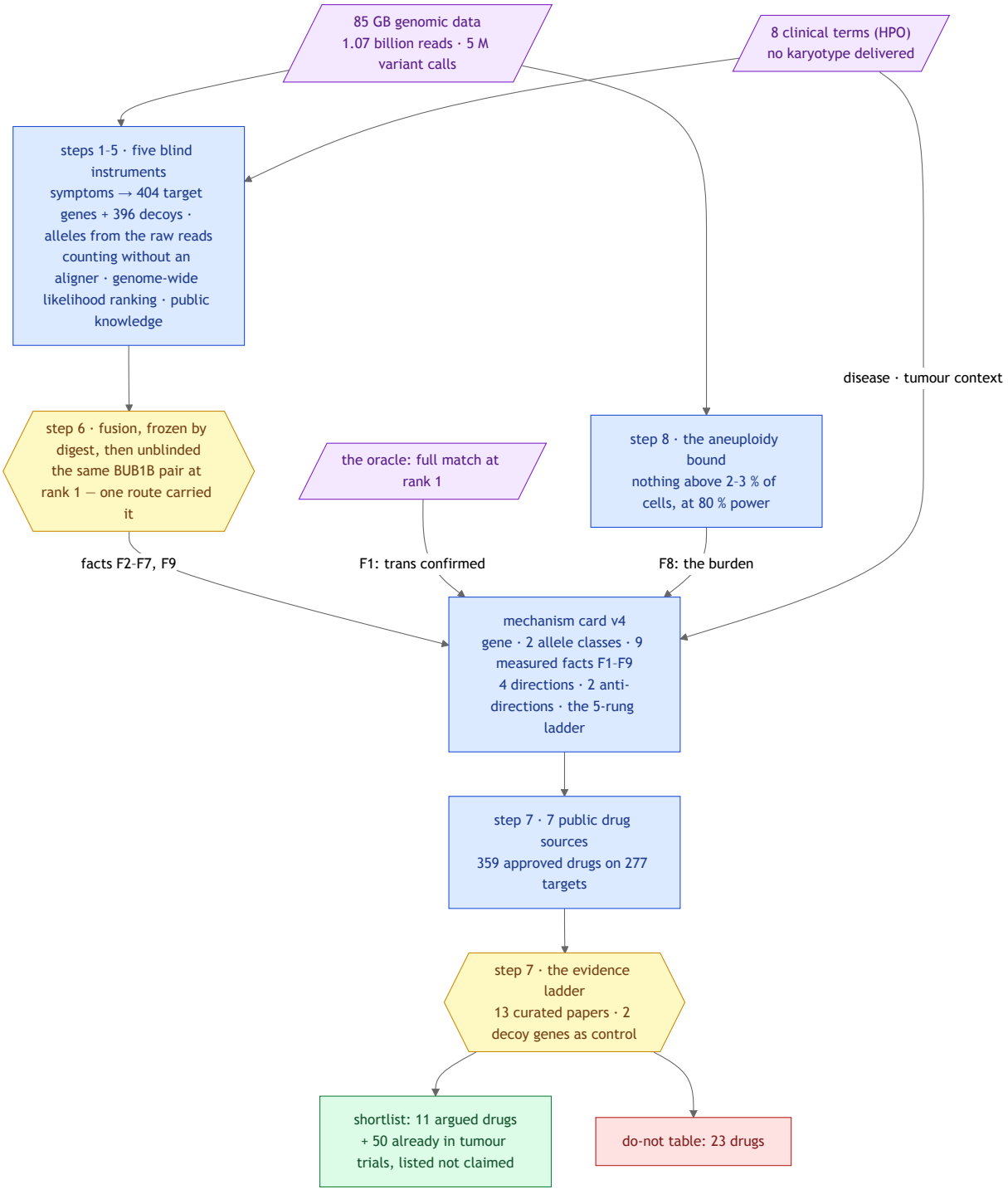
Direction	The idea	What we found	Standing after step 8
D1 reinforce the brake	strengthen the remaining checkpoint	**no approved drug does this**; every checkpoint drug on the market or in trials *weakens* it	the do-not table
D3 raise the amount of working BubR1	the weakened copy still works — make more of it last longer	**niacin / nicotinamide** (vitamin B3 forms): boosting the NAD^+ pathway raised BubR1 in mice bred with too little of it (rung 3, animal model, by proxy); valproic acid and vorinostat act on BubR1's acetylation switch (rung 2, class level)	**the deepest mechanistic lead** — acts on the constitutional genotype whatever the burden; thinnest automated support
D4 the child's actual tumour	approved, non-chemotherapy drugs with a mechanism in rhabdomyosarcoma	**statins**: rhabdomyosarcoma cells depend on making their own cholesterol; lovastatin itself slowed them (rung 2, molecule level); simvastatin has been in a completed paediatric sarcoma trial (NCT02390843)	**the best-supported lead** — a molecule-level result and a paediatric trial agree
D2 exploit the stress of aneuploid cells	aneuploid cells are already stressed; some approved drugs tip them over	**chloroquine** preferentially killed aneuploid cells in the reference screen (rung 2, molecule level); metformin, hydroxychloroquine, bortezomib inherit class results	**literature only** — the sequenced sample shows no aneuploid population ≥ 2–3 % for such a drug to act on; a tumour-context angle, not a treatment of the child's state

What not to do — the half a clinician may care about most. Thirteen approved drugs are *spindle poisons*: they jam cell division and rely on the checkpoint to hold the cell. In a child whose checkpoint is the broken part, that reliance is a hypothesis worth weighing. Vincristine and vinorelbine are standard rhabdomyosarcoma chemotherapy. **This report flags the question for the treating team; it does not advise withholding standard care.** No approved drug weakens the checkpoint further — that class exists only in trials, and it is the one to avoid.

What we do not know. Whether any drug on this page helps (rung 3 is one mouse paper). Whether the weakened BubR1 copy is unstable or simply inactive. What tissue was sequenced. Anything about the X and Y chromosomes, or about events smaller than a chromosome arm.

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2. Two datasets, one line of evidence — the whole path



> **Legend** — **purple = raw inputs** (the child's DNA, the child's clinical terms, the oracle's reply); **blue = machinery** (every box is a tool with a manifest and a methodology entry); **yellow = decision points** (the fusion that was frozen before it was compared; the ladder whose rungs were written before the search); **green = the goal**; **red = what to avoid**. The labelled arrows are the measured facts each step contributed to the card.

Both datasets act at every stage: the genomic data fixed the gene, the alleles, their dosage and the absence of aneuploidy in the sample; the clinical terms fixed the disease, the target set of step 1, the tumour context and — through the tumour — direction D4 and the paediatric questions. A result that uses only one says so.

3. How the genotype was established — steps 1 to 6 and 8, in the order they ran

Iteration 2 was designed as a **blind mirror** of the first answer: could the pair be found again from the raw material, by instruments sharing no code with iteration 1, with no iteration-1 table read until the last entry? It could — and the way it could is the finding.

Step	Question	What it measured	What it gave the drug argument
1 the blind target set	which genes do the eight symptoms alone nominate?	404 genes tiered by symptom evidence, 396 size-matched **decoy genes** beside them; held-out recall@300 **0.335** — a third of known disease genes would be missed. 73.6 % of protein-coding genes have no symptom curation at all	the decoy set that later controls the drug pipeline; the lesson that symptom-only ranking is an *enrichment*, never coverage
2 alleles from the raw reads	what do the reads carry in those genes, without the vendor's caller?	14.7 M reads recovered from 1.07 billion by a k-mer bait; two independent callers agreed with the vendor at 99.6 % / 99.3 %; **1,199 alleles the vendor never reported**; 164 structural variants, **none in the gene**; 6,823 phased blocks, none reaching the gene	F4: no hidden deletion; phase unmeasured in the reads — *trans* is the oracle's statement
3 counting, not aligning	are the alleles at one copy each, and every exon at two?	164,044 alleles typed from k-mer counts with no aligner and no caller; **both lead alleles at one copy** (posterior 1.0); every exon region at **two copies** (median 2.00; the X at 0.95 in a male, found with nothing told to the method)	F2, F3: a *null + weakened* genotype — some BubR1 exists to stabilise, so D3 is live
4 genome-wide likelihood ranking	with no hard filter, which pair has the highest posterior?	**51,920** biallelic hypotheses over 11,465 genes; `BUB1B` first at a ratio of 341,019; the missense's pathogenicity lane 4.33; targets separate from decoys at AUC 0.706; population phasing **undetermined**	F6, F7: "damaging" quantified; population data cannot corroborate *trans*
5 public knowledge	does the literature nominate the pair?	815 genes asked of LitVar, PubTator, UniProt/InterPro, AlphaFold and PubMed; the route separates from decoys (AUC 0.634) and is *not* citation volume; yet its top two genes are decoys and `BUB1B` sits at rank **293** of 815; the missense lies **outside every curated domain**, in a region the structure model is confident about (pLDDT 91)	F5: whether the weakened copy is unstable or inactive is **open**; D3 is argued as *more BubR1*, not *refold it*. The literature is searched per drug, never trusted as a ranker
6 fusion, then unblinding	fused, does the mirror rediscover the pair?	the ranking frozen (sha256 recorded) *before* any iteration-1 table was read; **the same pair at rank 1** under both weightings — but silencing the likelihood route drops it to rank 62; the symptom route had it at rank 9; the literature route never listed it. A second oracle upload: **0.000 bits** of information — not made	F9: a rediscovered, confirmed answer; and the honesty rule the drug section inherits — *one route found it, two corroborated at a distance*
8 the aneuploidy bound	is mosaic aneuploidy detectable in the sample, and at what limit?	2,778,844 heterozygous calls reduced to 127,913 arm rows; a fold-over allele-balance statistic reaches 80 % power at **2 % of cells** (chr1–6) and **3 %** (chr7–22), where the statistic iteration 1 used had no power at any fraction; **no autosome above its own limit**; the model family for *which* trisomy does not fit and no mechanism is named; the divisor holds under three injected abnormal chromosomes	F8: **measured** — nothing above 2–3 % in the sequenced sample; D2 becomes literature-only

Three things this path teaches that shape the drug section: the decoys were the most useful thing built (they later exposed a non-specific drug direction); a "negative" result is only one when it carries its detection limit

(step 8 corrected step 0's flat karyogram on exactly that point); and a rank-1 answer with one route carrying it is reported as such, not as a consensus.

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4. The mechanism (methods form Q14: loss- versus gain-of-function, pathway, consequence)

4.1 Loss of function, of a specific shape

copy	what changed	what it does to the protein	evidence
one	a *stop* change in exon 17 of 23	the protein ends early, before its last working part — **no functional protein**	ClinVar VCV000533901, pathogenic / likely pathogenic for MVA1
the other	a *missense* change in exon 23 of 23 — one amino acid swapped	a **weakened** protein: never seen in 140,000 healthy genomes, predicted damaging by two independent predictors; outside every catalogued domain, in a region the structure model is confident about	iteration 1 entries 070–074; iteration 2 entries 099, 106

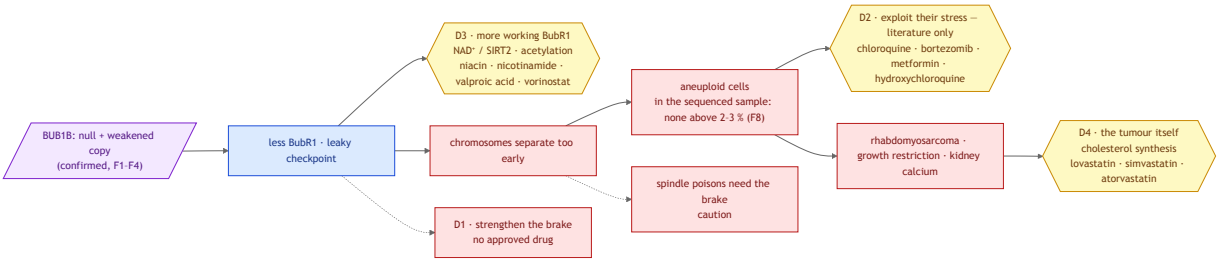
No gain-of-function or dominant-negative activity is reported for `BUB1B`; complete loss is lethal before birth in mouse and human (Hanks 2004, PMID 15475955; Suijkerbuijk 2010, PMID 20516114), which is why "switch BubR1 off" is never a therapeutic direction.

4.2 The nine measured facts on the card (version 4)

fact	measured	what it changes for a drug
F1 the two copies are on different chromosomes (*trans*)	**confirmed** by the answer key (full match, rank 1)	the argument stands on a confirmed genotype
F2 each copy present once in the raw reads	yes — both alleles at one-copy dosage, no aligner	a *null + weakened* genotype: some BubR1 exists (D3 live)
F3 no deleted or duplicated stretch in the gene	every exon at two copies	the two alleles are the whole story
F4 structural variants and read-backed phase	none in the gene; phase unmeasured in the reads	*trans* is the oracle's statement, and the report says so
F5 where the missense sits	outside every catalogued domain, pLDDT 91	unstable or inactive: **open**; D3 is a dosage argument
F6 how damaging the missense is	AlphaMissense likely-pathogenic (lane 4.33); the pair leads 51,920 hypotheses	"damaging" quantified
F7 population phasing	undetermined — 5 of 50 leading alleles reach gnomAD	population data cannot corroborate F1
F8 aneuploid fraction in the sequenced sample	**no autosome above its limit: 2 % (chr1–6), 3 % (chr7–22) at 80 % power**; X/Y and sub-arm events untestable by this route; tissue unnamed	**D2 cannot be argued from a burden in this child**; D3, D4 and the do-not table are untouched
F9 the blind mirror	found the same pair first	a rediscovered, confirmed answer

4.3 The pathway, and where a drug could stand

BubR1 is a core part of the **spindle assembly checkpoint** (SAC — the wait-until-attached brake) and of the complex that switches off the machinery pulling chromosomes apart. Less BubR1 means a **leaky brake**: division can proceed with chromosomes still unattached.



> **Legend** — **purple** the genotype; **blue** the molecular consequence; **yellow** the three places an approved drug could stand, each a direction below; **red** what goes wrong, and the two dotted boxes are the direction with no drug and the class to be cautious with.

Downstream, in order: premature separation of chromosomes → the aneuploidy that names the disease (not detectable above 2–3 % in the sample sequenced here) → in surviving aneuploid cells a chronic stress from carrying the wrong number of copies of many proteins → a predisposition to exactly the cancers this child has → growth restriction and nephrocalcinosis, the delivered clinical terms.

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5. How the candidates were found, graded and controlled

5.1 One card, seven public sources, no hand-typed query

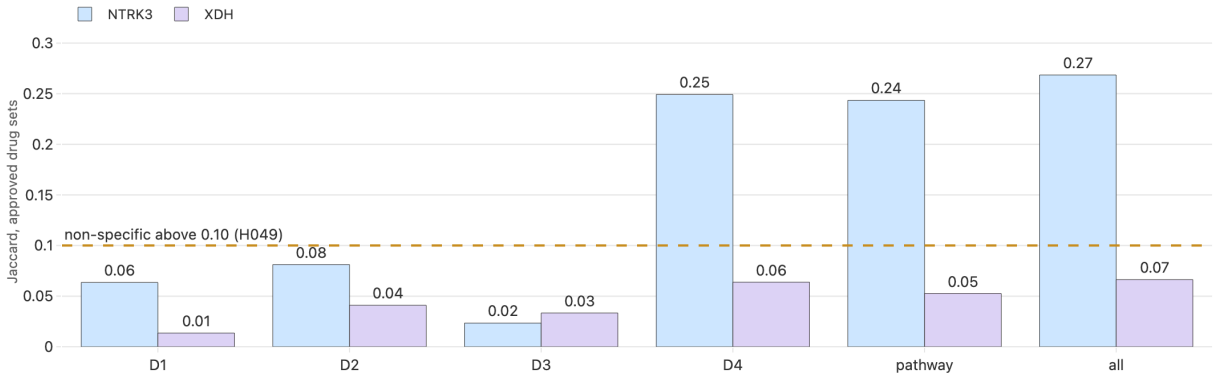
The mechanism card drives every query. Reactome widens the card's 32 seed genes to 277 targets; Open Targets, ChEMBL and DGIdb return every drug known to act on them (3,859 rows, **359 market-approved drugs**); openFDA confirms a US label and whether it has a *Pediatric Use* section; ClinicalTrials.gov answers "in this tumour?" and "in children, anywhere?"; PubMed answers "how much has been written about this drug with this mechanism?" — per drug, with the vocabulary read off the card. Every table carries a manifest with the retrieval time and a digest of every reply; the same commands run another case by swapping the card.

5.2 The ladder, written before the search

rung	meaning	how it is tested
5	approved for a related indication, with paediatric use	approved **and** a registered trial in the child's tumour **and** a paediatric pointer (label section 8.4, or a study admitting children) — **applied to the tumour direction only**
4	clinical trial in a related indication	a registered trial in the child's tumour — tumour direction only
3	animal model of the mechanism	a paper read by a person, with its PubMed id
2	cell model of the mechanism	a paper read by a person, with its PubMed id
1	mechanism only	acts on a target in the direction's set — *named, never submitted*

Automated and curated tests print side by side. Inherited evidence is marked `proxy` with the compound actually tested. **One rule was narrowed after the first run and the change is recorded:** a rhabdomyosarcoma trial was lifting drugs onto rung 5 on the *mechanism* directions (85 drugs, tetracycline among them); a trial in the tumour says nothing about a checkpoint, so trial-based rungs now grade the tumour direction alone (entry 113). A second pre-registered test (entry 120) then showed that adding the measured aneuploid fraction to the card moved **no rung at all** — the ladder is independent of F8 by construction, and the test caught a file-reading fault along the way that would otherwise have gone unnoticed.

5.3 The control: the same child, a different gene

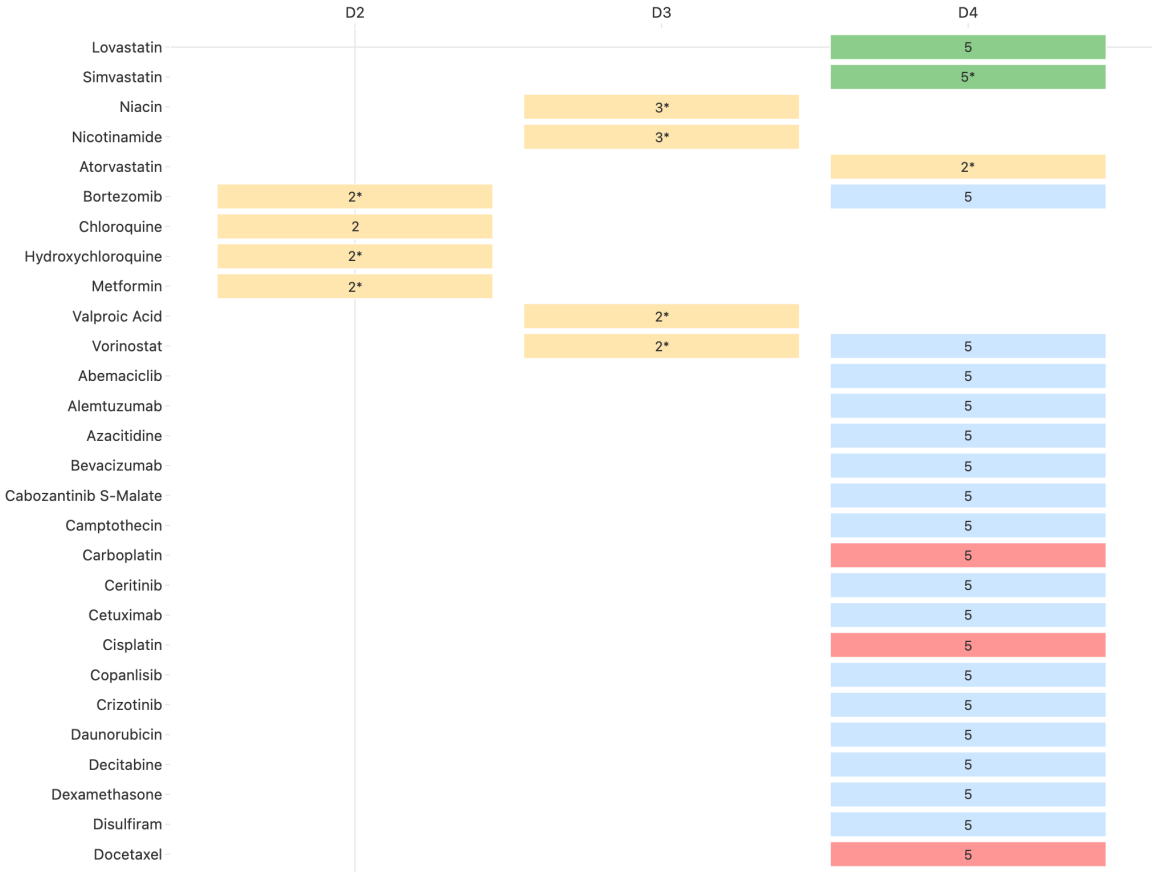


> **Legend** — one bar per decoy gene; the dashed yellow line is the pre-registered 0.10 threshold. **D4 is above it** against NTRK3 (Jaccard 0.25): the cancer pharmacopoeia comes out for any cancer-adjacent gene, and 64 of the decoy's drugs had a rhabdomyosarcoma trial with the gene never consulted. **D1–D3 are below it** against both decoys. Consequence: D4 is argued only from curated mechanism results, never from "this drug has a rhabdomyosarcoma trial".

The decoys were chosen before the run: NTRK3 by a seeded draw from step 1's 396 matched decoys (a receptor kinase with approved inhibitors — a harder control than a random gene), and XDH, a metabolic enzyme with nothing to do with cell division.

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6. The candidates



> **Legend** — the number is the rung; **green** a curated result and rung 4–5; **yellow** a curated result at rung 2–3; **blue** the automated tests alone (already being tried in the tumour — not a finding of this work); **red** set aside; `*` inherited (proxy) evidence.

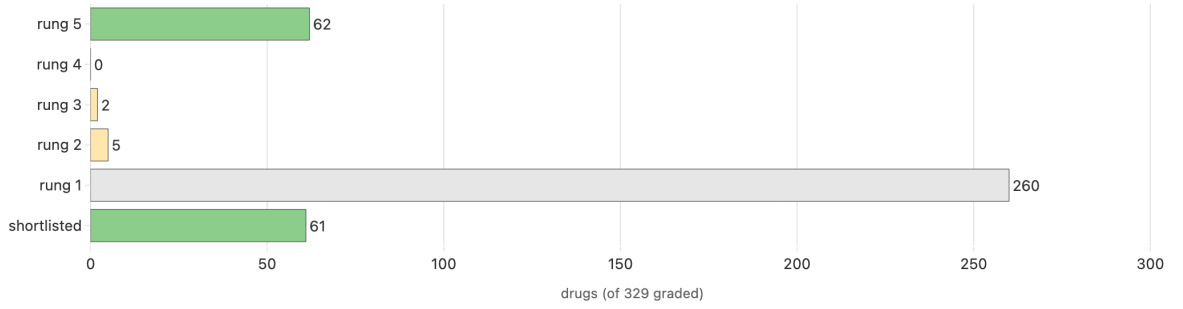
6.1 The eleven drugs with a cited mechanism result, ordered as this report argues them

#	drug (approved for)	direction	rung	the cited result	inherited?	given to children?
1	**lovastatin** (cholesterol)	D4 tumour	5	lovastatin itself slowed rhabdomyosarcoma cells; the tumour cannot take up cholesterol instead of making it (Gizaw 2025, PMID 41249736); a completed phase-1 study gave a statin with chemotherapy to children with relapsed solid tumours including rhabdomyosarcoma (NCT02390843)	no	label 8.4; studies admitting children
2	**simvastatin** (cholesterol)	D4	5	same class result; it is the statin in NCT02390843	yes (lovastatin)	studies admitting children
3	**atorvastatin** (cholesterol)	D4	2	same class result; carries a paediatric indication on its label	yes (lovastatin)	label 8.4; studies
4	**nicotinamide** (vitamin B3)	D3 more BubR1	3	the NAD^+ precursor NMN raised BubR1 in mice; more SIRT2 extended the life of BubR1-deficient mice (North 2014, PMID 24825348)	yes (NMN)	studies admitting children
5	**niacin** (cholesterol; vitamin B3)	D3	3	same result; the prescription-strength NAD^+ precursor	yes (NMN)	studies admitting children
6	**valproic acid** (epilepsy)	D3	2	HDAC2/3 strip the acetyl mark that protects BubR1 from degradation (Park 2017, PMID 28985013; Choi 2009, PMID 19407811) — an HDAC inhibitor would be expected to slow that loss; inferred, not shown	yes (class)	label 8.4; studies
7	**vorinostat** (lymphoma)	D3	2	same class mechanism	yes (class)	label 8.4; studies
8	**chloroquine** (malaria)	D2 stress — literature only	2	chloroquine itself preferentially killed cells with an extra chromosome (Tang & Amon 2011, PMID 21315436)	**no**	studies admitting children
9	**bortezomib** (myeloma)	D2	2	aneuploid human cells depend on faster protein degradation and are more sensitive to proteasome inhibitors (Ippolito 2024, PMID 39247952; Chui 2019, PMID 31530568)	yes (class)	label 8.4; studies
10	**metformin** (diabetes)	D2	2	the AMPK activator AICAR was aneuploidy-selective in the same screen	yes (AICAR)	label 8.4; studies
11	**hydroxychloroquine** (lupus)	D2	2	chloroquine's result, inherited by its clinical analogue	yes (chloroquine)	label 8.4; studies

Why this order and not the ladder's: the ladder ranks by rung, and by rung the statins lead. The report then places D3 ahead of D2 because **D3 acts on the constitutional genotype the child has whatever the burden**, while D2's whole rationale is a population of aneuploid cells that step 8 could not find above 2–3 %

in the sample sequenced. Within D2, chloroquine keeps its molecule-level result; the direction as a whole is a tumour-context and chemoprevention hypothesis argued from published biology, and the report says so wherever it is quoted.

Reading the table honestly: **no** drug has a molecule-level animal result on the checkpoint itself. The deepest mechanistic lead rests on one mouse paper, inherited by two vitamin-B3 forms. The best-supported tumour lead has a completed paediatric trial and a molecule-level cell result. The only molecule-level aneuploidy result is chloroquine's, and it now has no measured target population in this sample.



> **Legend** — drugs counted once at their best rung; **green** rung 4–5 and the shortlist, **yellow** rungs 2–3, **grey** rung 1. Most of the 62 at rung 5 are there by the automated test alone: approved, trialled in rhabdomyosarcoma, used in children.

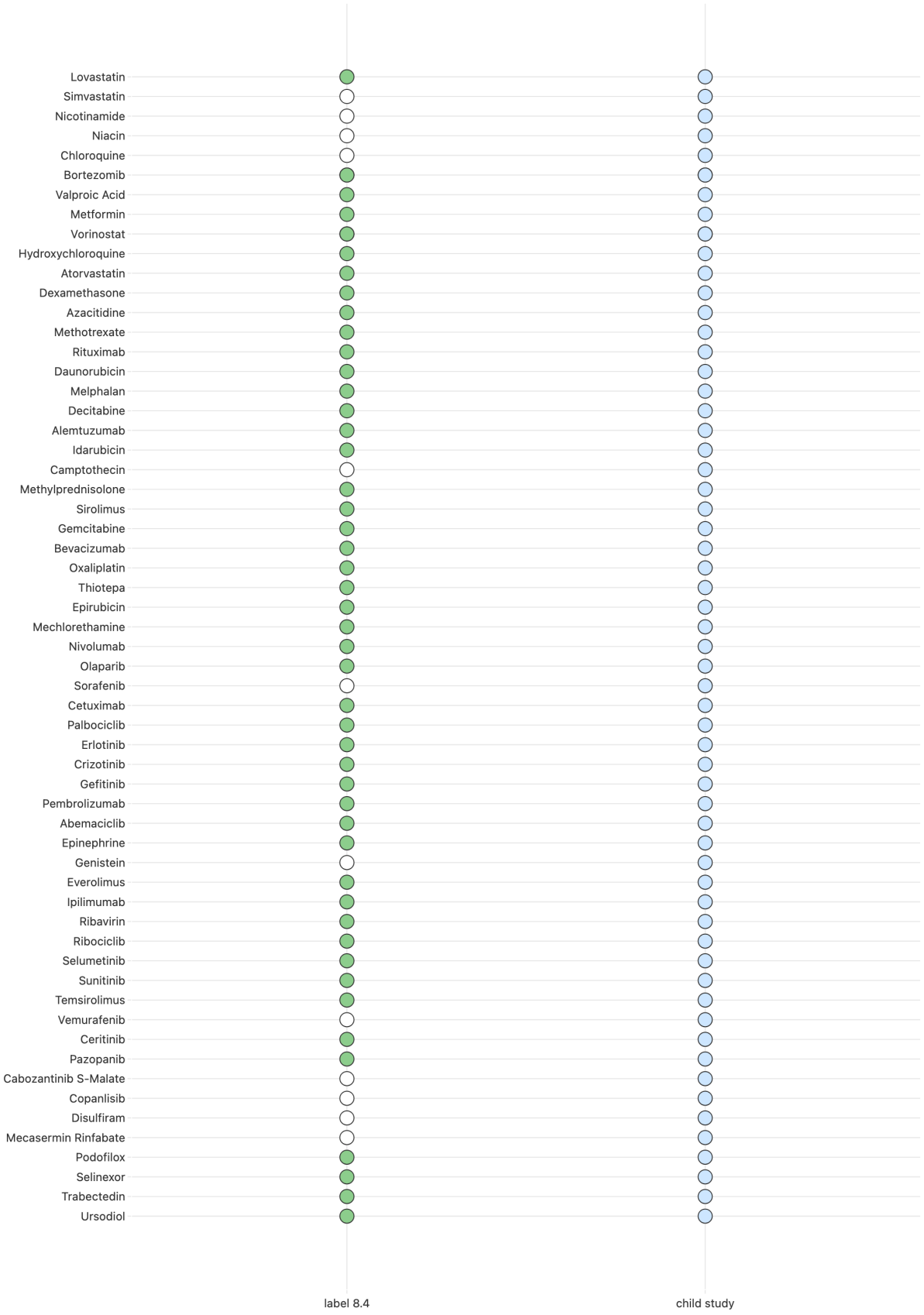
6.2 Fifty drugs already being tried in rhabdomyosarcoma — listed, not claimed

Fifty more approved drugs reach rung 5 automatically (dexamethasone, azacitidine, methotrexate, bevacizumab, the CDK4/6 and mTOR inhibitors, the checkpoint antibodies…). They are in `shortlist.tsv` with the verdict *context trial only – not a mechanism finding*, because the decoy control showed the same list comes out for a gene unrelated to the checkpoint. They are the tumour's trial landscape; a reader deserves to see them; they are not this work's finding.

6.3 The do-not table

verdict	drugs	why
caution — spindle poisons (13)	vincristine, vinorelbine, vinblastine, vinflunine, paclitaxel, docetaxel, cabazitaxel, eribulin, ixabepilone, epothilone D, colchicine, brentuximab vedotin, polatuzumab vedotin	these drugs *rely on the checkpoint to hold the cell*; MVA cells have a weakened checkpoint and slip through (Suijkerbuijk 2010, PMID 20516114; Sinha 2019, PMID 30601084). Vincristine and vinorelbine are standard rhabdomyosarcoma chemotherapy: **a hypothesis for the treating team to weigh, never advice to withhold treatment** — a regimen with survival data behind it outweighs a theoretical mechanism concern
do not — further checkpoint weakening (0 approved)	Mps1/TTK and Aurora inhibitors	aneuploid cancer cells are selectively killed by weakening the checkpoint further (Cohen-Sharir 2021, PMID 33505028) — the same logic makes a child whose checkpoint is constitutionally weak the worst candidate. None is approved; every one in the candidate table is phase 3 or earlier
standard of care (10)	vincristine, dactinomycin, cyclophosphamide, ifosfamide, doxorubicin, etoposide, irinotecan, temozolomide, carboplatin, cisplatin, topotecan	already the treatment for the tumour; giving them is not a repurposing. Named so a reader sees they were considered, not missed

6.4 Has each candidate been given to children?



> **Legend** — **green** a US label with a *Pediatric Use* section (8.4); **blue** at least one registered study admitting children, in any indication; empty = neither. Every argued drug has at least one pointer. **A pointer is something to read, never a licence.**

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7. What this report cannot tell you

1. **Whether any of these drugs helps this child.** Rung 3 is one mouse paper; rung 2 is a dish. Hypotheses for follow-up.
2. **What tissue was sequenced, and therefore what F8 means for the diagnosis.** The delivery does not name it; a karyotype's fraction is measured in cultured cells. F8 says the sequenced sample carries no mosaic population above 2–3 % on any autosome; it says nothing about X, Y, or events smaller than a chromosome arm, and the dosage route that could is not yet run (step 8, H054 open).
3. **Whether the weakened BubR1 copy is unstable or inactive.** Outside every catalogued domain; the structure model is confident about the region but silent on function.
4. **Anything about a drug the public sources never listed** beyond what a person put in the curated table — four of the eleven (chloroquine, hydroxychloroquine, niacin, nicotinamide) are there only because a person read the paper; their rows say `curated_only`.
5. **The curated rungs are one reader's**, thirteen abstracts on one day; every row carries its PubMed id so it can be checked.
6. **DrugBank was not queried** (licence); Open Targets, ChEMBL, DGIdb, openFDA, Reactome, ClinicalTrials.gov and PubMed were.

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8. Reproducibility and cost

Every step is one command, case-agnostic, with a manifest and a lineage row; no patient data appears in any Track 2 artefact (gene symbols, allele *classes*, drug names, trial ids, PubMed ids only):

```
 uv run mva-drugs snapshot                          # card → 5 tables, 7 public APIs, cached repl
uv run python tools/track2_decoys.py ...           # a decoy card and its snapshot (×2)
uv run python tools/track2_enrich.py               # literature, paediatric trials, labels, ladde
uv run python tools/render_track2_figures.py       # the figures in this report
uv run python tools/record_track2_lineage.py       # the lineage edges into the warehouse ledger
```

Two laptop runs of about ten minutes each against public APIs, seconds when cached; the genotype steps 1–6 and 8 ran on a laptop and one Linux server over a fortnight (their methodology entries state each cost). Code and tables: the public repository named in the form; the dashboard's `/track2` page renders the same tables with the grading visible.

9. Where every claim lives

claim	register id	verdict · confidence	entry
D1 yields no submittable drug and becomes the do-not table	H045	supported · 0.80	113
D2 holds an approved compound with a molecule-level cell result	H046	supported · 0.70	113
D3 has an approved route on the NAD^+/SIRT2 axis with animal evidence, by proxy	H047	supported · 0.60	113
D4 holds a non-chemotherapy approved drug at rung ≥ 2 with paediatric data	H048	supported · 0.80	113
the automated stages are non-specific (D4 overlaps a decoy ≥ 0.10)	H049	supported · 0.85	112
every argued drug has a paediatric pointer	H050	supported · 0.70	113
the ladder is independent of the measured aneuploid fraction (card v4 moved no rung)	H061	supported · 0.85	120
no chromosome carries a mosaic aneuploidy above its own detection limit	H051 (step 8)	supported · 0.80	116
the detection limit is ≤ 10 % of cells for every autosome (measured 2–3 %)	H052 (step 8)	supported · 0.85	116
the blind fused ranking put the confirmed pair first	H041/H043 (step 6)	see entries	108–109
no approved drug acts directly on BubR1	H011 (iteration 1)	supported · 0.80	078

Confidence is a transparent triage heuristic printed with its arithmetic, never a probability. The register is append-only and shows every revision, including two committed too early and corrected in the open (entries 113 and 120).